Shoumik Saha

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RESEARCH INTEREST

- **Machine Learning** - Adversarial Machine Learning, Security for Machine Learning
- **Security** - Computer security, Malware, Adversarial Attack

PROFESSIONAL EXPERIENCE

Graduate Research Assistant **Full Time**
Department of Computer Science *Aug 2022 – Present*
University of Maryland

Supervisor - Dr. Tudor Dumitras

Lecturer **Full Time**
Department of Computer Science and Engineering *Jul 2021 – Jul 2022*
United International University

Research Assistant **Part Time**
Data Science and Engineering Research Lab *Mar 2021 – Jul 2022*
Bangladesh University of Engineering and Technology (BUET)

Under the supervision of
Dr. Mohammed Eunus Ali, Dr. Atif Hasan Rahman

EDUCATION

University of Maryland
Ph.D. in Computer Science and Engineering (Ongoing) *Aug 2022 – Present*

- CGPA – *3.90/4.0*
- Advisor – Dr. Tudor Dumitras

Bangladesh University of Engineering and Technology (BUET)
BSc. in Computer Science and Engineering *Mar 2016 – Feb 2021*

- CGPA – *3.66/4.0*
- Advanced GPA – *3.79/4.0*
- Thesis supervisor – Dr. Atif Hasan Rahman

Notre Dame College
Higher Secondary Certificate *Jul 2013 – Jun 2015*

- GPA – 5.00/5.00

RESEARCH

- **Shoumik Saha**, Sadia Afroz, Atif Rahman.
  MALIGN: Adversarially Robust Explainable Malware Family Analysis using Sequence Alignment (BSc Thesis) (pdf)
  A novel approach that incorporated ideas and tools from Bioinformatics into Malware Security. We developed a static raw-byte-based malware family classifier with better accuracy and robustness. Malign also provides explainability by relocating the code blocks responsible for malicious attacks.
  Status: Under Review
- **Shoumik Saha**, Wenxiao Wang, Yigitcan Kaya, Soheil Feizi, Tudor Dumitras.
  Adversarial Robustness of Learning-based Static Malware Classifiers (arxiv) (pdf)
  We explored Universal Append Patch (UAP) attack on byte-based static malware classifier for the first time. Then we applied de-randomized smoothing on our defense model to make it robust against such attack.
  Status: Ongoing

- Yigitcan Kaya, Yizheng Chen, **Shoumik Saha**, Fabio Pierazzi, Lorenzo Cavallaro, David Wagner, Tudor Dumitras
  Is Machine Learning Sufficient for Malware Detection in the Wild? (pdf)
  We presented a quantitative study of how sandbox traces differ from real-world ones, and how it impacts machine learning models. We identified this distribution shift and proposed a solution for ML models that boosted the TPR from 14% to 20%@1%FPR.
  Status: Will be submitted to IEEE S&P 2024
- **Shoumik Saha**, Subangkar Shanto, Atif Rahman, Mohammad Masud, Mohammed Eunus Ali
  Contrastive Self-Supervised Learning Based Approach for Patient Similarity: A Case Study on Atrial Fibrillation Detection from PPG Signal (pdf)
  We proposed Contrastive Learning based Deep Learning solution to find the patient similarity using physiological signal. We demonstrated the efficacy of our method by applying it to detect Atrial Fibrillation using PPG signals collected from smartwatches.
  Status: Under Review

SCHOLARSHIPS AND ACHIEVEMENTS

- Awarded *'Dean's Fellowship'* from the University of Maryland
- Awarded *'Innovation Fund'* for research from the ICT division of Bangladesh government
- Achieved *'Merit Stipends'* from BUET in five out of seven terms
- Got *'Dean's Award'* in Junior year for extra-ordinary result
- Achieved *'Talent-Pool Scholarship'* in High School

SELECTED ACADEMIC PROJECTS

- **Website Development for Flight and Hotel Room Reservation System** (Github link)
  Developed a website where customers can book flight tickets, hotel rooms through card payment, and companies can update their page and manage bookings. *Tools:* Django, SQLite
- **Software Development for Football Club Management** (Github link)
  This application stores all information about the club (players, managers, games). Admin of the club can update, add and delete these information and generate graphs from statistics. *Tools:* Java, JavaFX, Oracle Database
- **Micro-controller Project for Real-time Detection of Car Theft** (Youtube video link)
  Can detect a car in real time and send a warning message with gps location to the owner's mobile. *Tools:* Arduino, GPS, GSM module, Sonar, LDR sensor
- **Multi-player Shooting Game** (Github link)
  A 2nd year project, two players can connect from two different machines under the same network. *Tools:* Java, JavaFX, Java Networking
- **Tic-Tac-Toe** (Github link)
  A 1st year project, single player Tic-Tac-Toe game with Mini-Max algorithm. *Tools:* C, iGraphics
- **Miscellaneous**
  - Developing a C compiler using lexical analyzer and parser designing tools
  - Designing a 4-bit Computer Model using Atmel Studio, MIPS architecture
  - Simulating Mancala game in AI lab course
  - Implementing and modifying some functionalities of XV6 Operating System
  - Modifying some functionalities of Computer Network in NS2.

ACTIVITIES

- General Secretary, BUET Photographic Society
- President, Notre Dame Nature Study Society
- Participant, ICPC Dhaka Regional
- Vice President, Gregorian Science Club

STANDARDIZED TEST SCORES

- *TOEFL:* 111/120 (R: 28, L: 28, S: 27, W: 28)
- *GRE:* 321/340 (Quant: 167, Verbal: 154)